GP-127

Adaptive Edge Vision For Distributed Eye Care :

A Frugal AI Framework for Glaucoma Diagnosis in Low-Resource Settings

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INTRODUCTION

- Gaucoma is the leading cause of irreversible blindness with 80M people affected as of 2020, and projected to reach 111M people by 2040, (Glaucoma Research Foundation)
- It is often referred to as “the silent thief of sight”, it causes progressive, permanent optic nerve damage, starting with asymptomatic vision loss.

PROBLEM STATEMENT

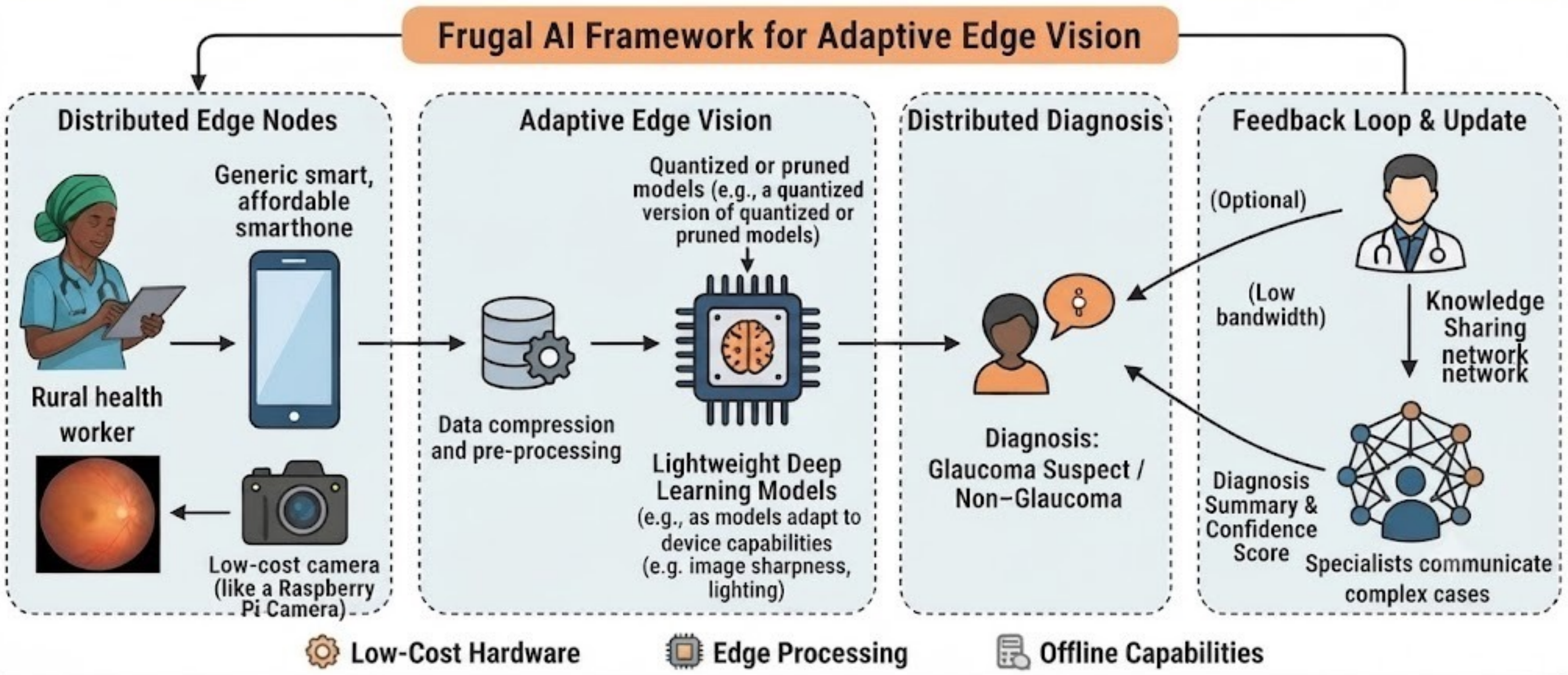
Rural Clinics Often Encounter:

- **Scarcity of Specialists:** High patient-to-opthalmologist ratios.
- **Infrastructure Gaps:** Limited access to expensive imaging e.g OCT - Optical Coherence Tomography OCT
- **Bandwidth Constraints:** Heavy cloud based AI models are impractical due to unstable internet.

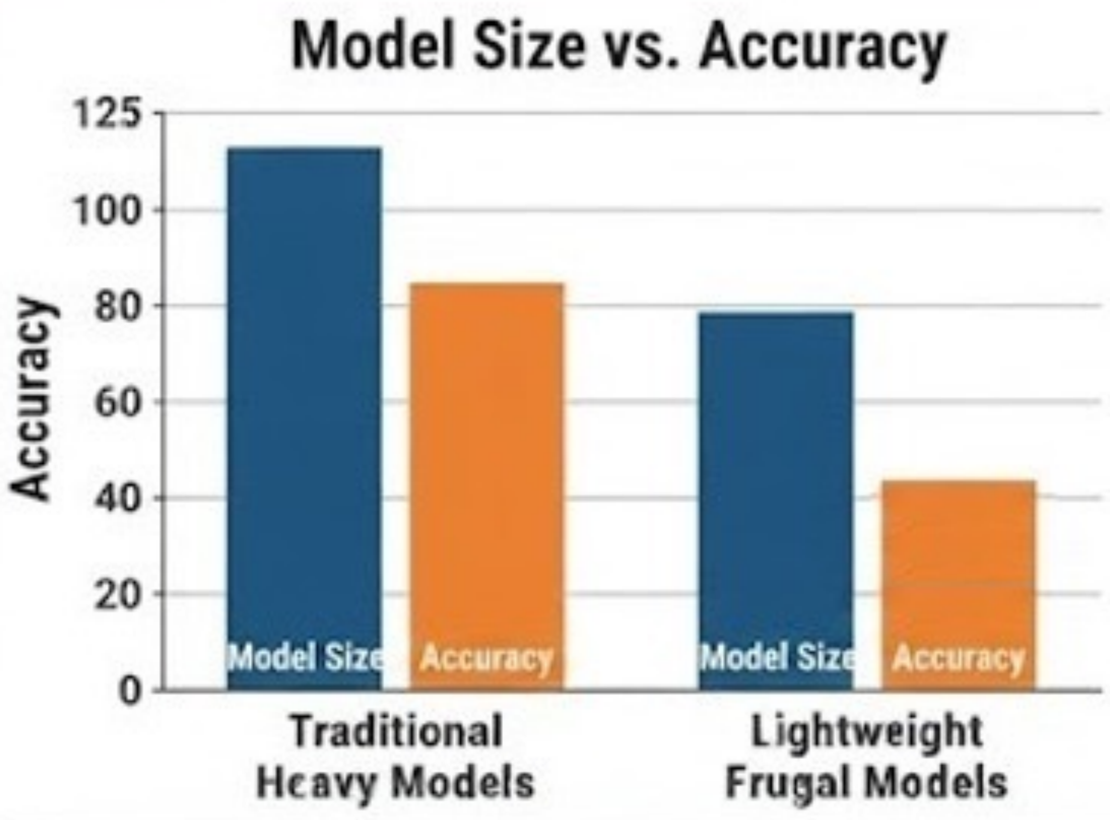
RESEARCH GAP

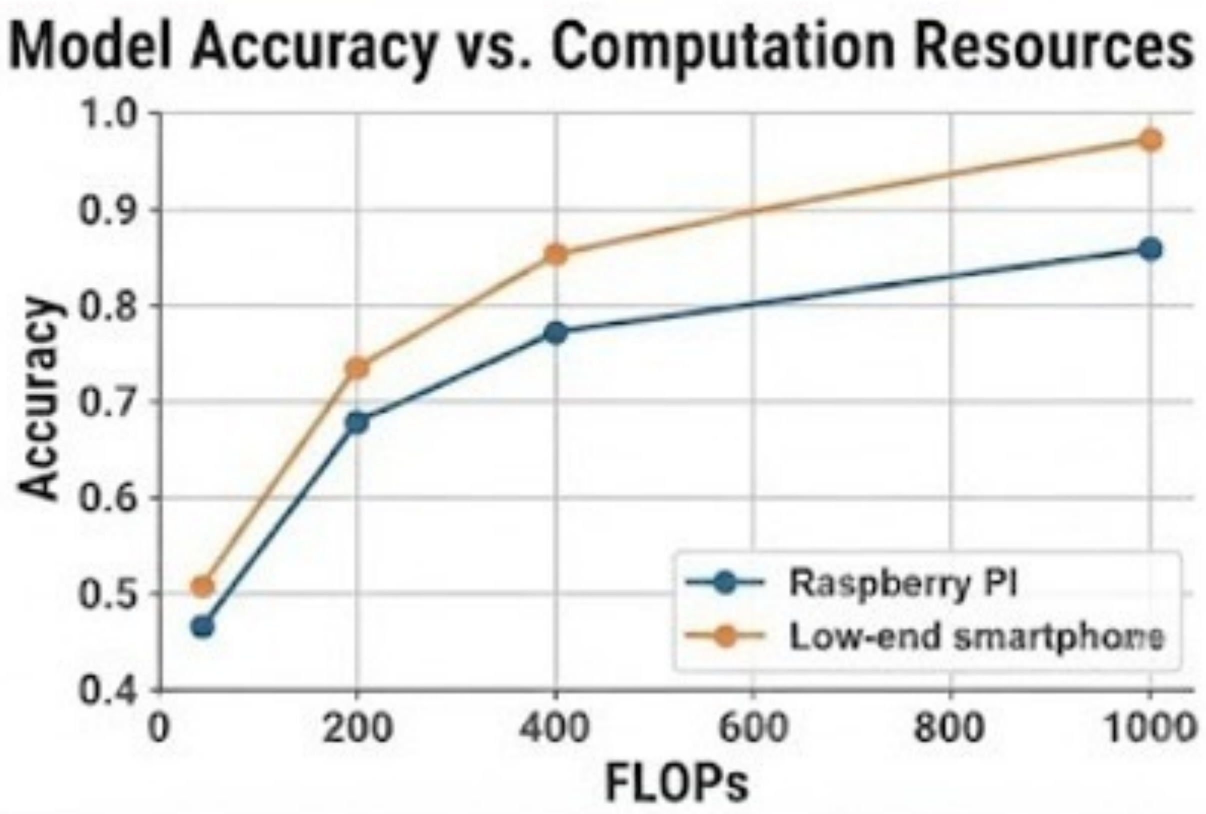
- Most AI models (e.g. Vision Transformers) are too ‘heavy’ for $200 smartphones or Rasberry Pi-based cameras affordable for remote areas.
- The vision is to create light weight AI models that can run on smartphones, or use cheaper cameras

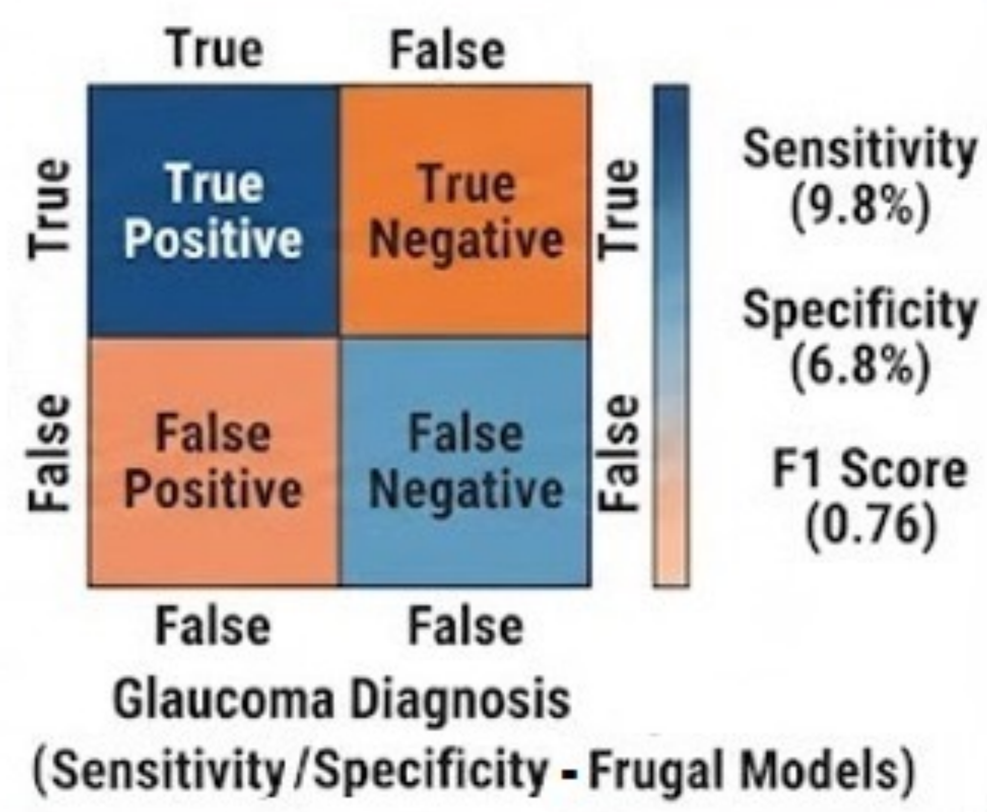
PROPOSED SOLUTION METHODOLOGY



RESEARCH FINDINGS







CONCLUSION

- Developed a frugal AI framework for glaucoma diagnosis diagnosis using lightweight deep learning model working on edge devices
- Demonstrated high performance and robustness in resource-constrained environments
- Significantly reduces reliance on expensive equipment, expert availability, and stable internet
- Enables scalable, accessible distributed eye care in low-resource settings

Kaggle Glaucoma Dataset, 3698 images

REFERENCES

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